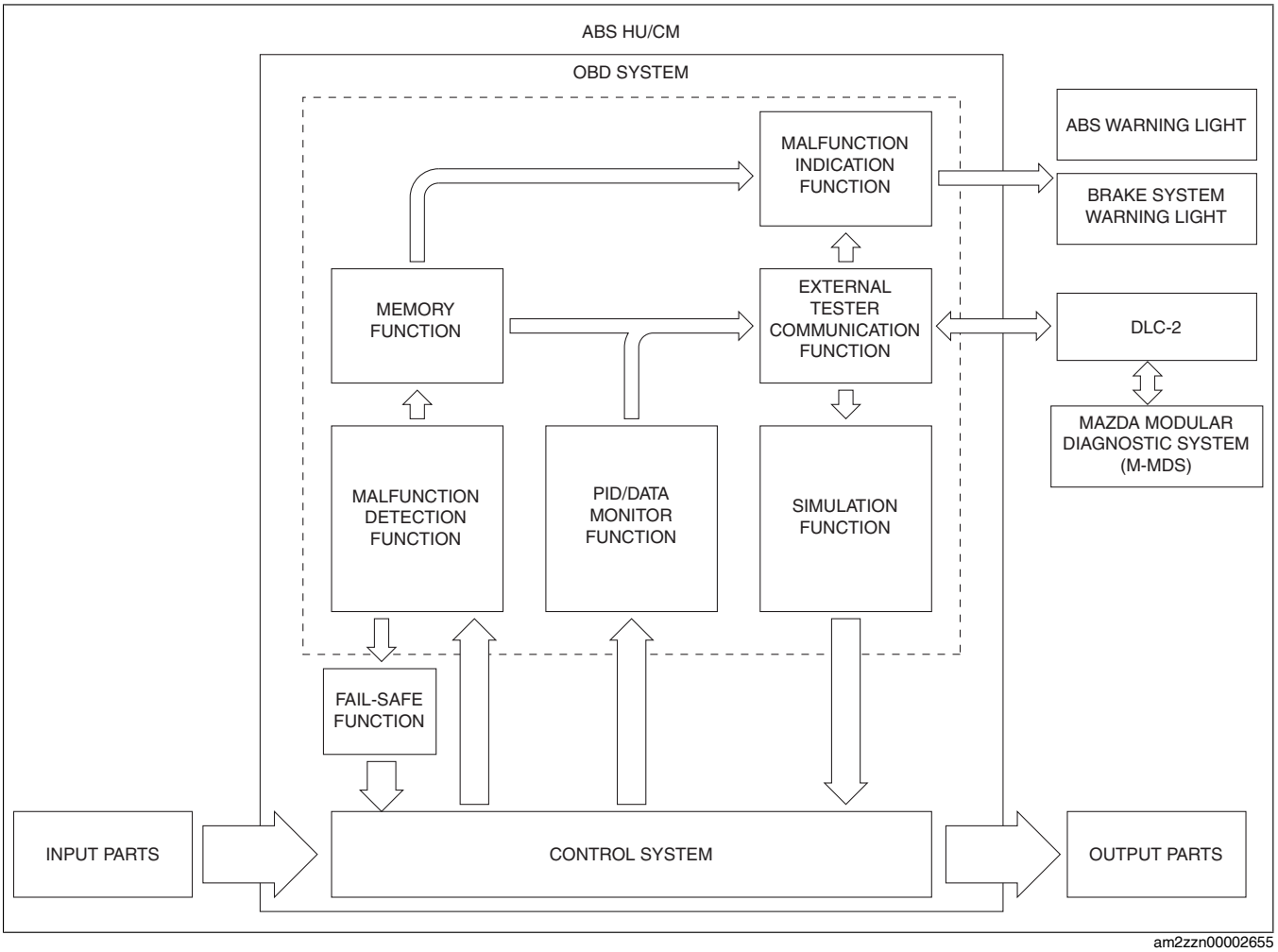
ON-BOARD DIAGNOSTIC SYSTEM [ABS HU/CM]

id040260181300

Outline

- The on-board diagnostic system consists of the following functions:
  - A malfunction detection function that detects abnormalities in input/output signals when the ignition switch is at the ON position.
  - A PID/data monitor function that reads out specified input/output signals.
  - A simulation function that allows for override operation of output parts (such as solenoid valves).
- The data link connector 2 (DLC-2), which groups together all the connectors used for malfunction diagnosis into a single location, has been adopted, thereby improving serviceability. Diagnosis is performed by connecting the Mazda Modular Diagnostic System (M-MDS) to the DLC-2.
- In addition to DTC read-out, the Mazda Modular Diagnostic System (M-MDS) has following functions:
  - PID/data monitor access
  - DTC clear using the display screen of the diagnostic tester

Block diagram



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Function

Malfunction detection function

- The malfunction detection function detects malfunctions in the input/output signal system of the ABS HU/CM when the ignition switched ON or driving the vehicle.
- The ABS HU/CM performs the following malfunction detections:
  - The ABS warning light and brake system warning light illuminate for approx. 3.0 s when the ignition is switched ON. At the same time, the input/output signals of each part are monitored for malfunction diagnosis.
  - After starting to drive, the first time the vehicle speed is approx. 10 km/h {6.2 mph} or more the pump motor is operated and malfunction diagnosis is performed again. The input/output signals of each part are then constantly monitored and malfunctions are diagnosed.

Note

- The sound of the pump motor operation can be heard.

Malfunction display function

- When the malfunction detection function detects a malfunction, the corresponding lights illuminate to alert the driver.
- Using the external tester communication function, DTCs can be output to the DLC-2 via the CAN line. At the same time, malfunction detection results are sent to the memory function and fail-safe function.

Memory function

- The memory function stores DTCs of malfunctions in input/output signal systems. With this function, once a DTC is stored it is not cleared after the ignition switch has been switched OFF, even if the malfunctioning signal system has returned to normal.
- Since the ABS HU/CM has a built-in non-volatile memory, DTCs are not cleared even if the battery is removed. Therefore, it is necessary to clear the memory after performing repairs. Refer to the Workshop Manual for the DTC clearing procedure.

DTC table

×: Applicable
—: Not applicable

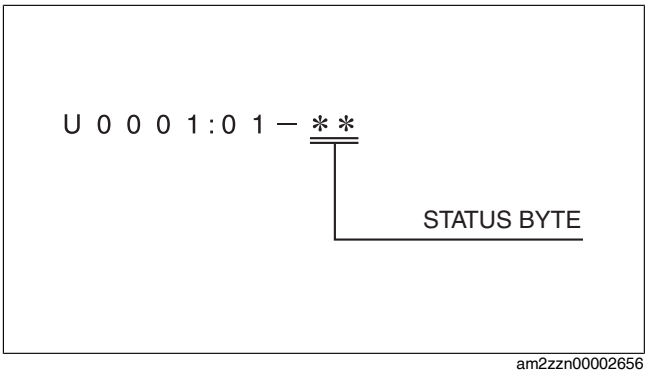
DTC	ABS warning light	Brake system warning light (when parking brake is released)	Description	Fail-safe	Drive cycle	Self test type*1	Memory function
C0010:01	Illuminates	Illuminates	ABS HU/CM internal malfunction (solenoid valve system)	×	—	C, D	×
C0011:01							
C0014:01							
C0015:01							
C0018:01							
C0019:01							
C001C:01							
C001D:01							
C0020:11	Illuminates	Not illuminate	Pump motor, motor relay	×	—	C, D	×
C0020:12							
C0020:13							
C0020:71							
C0030:07	Illuminates	Not illuminate*2	LF ABS sensor rotor	×	—	C, D	×
C0031:07			LF ABS wheel-speed sensor/ABS sensor rotor				
C0031:11	Illuminates	Not illuminate*2	LF ABS wheel-speed sensor	×	—	C, D	×
C0031:15							
C0031:29	Illuminates	Not illuminate*2	LF ABS wheel-speed sensor/ABS sensor rotor	×	—	C, D	×
C0031:2F							
C0031:64							
C0033:07	Illuminates	Not illuminate*2	RF ABS sensor rotor	×	—	C, D	×
C0034:07			RF ABS wheel-speed sensor/ABS sensor rotor				
C0034:11	Illuminates	Not illuminate*2	RF ABS wheel-speed sensor	×	—	C, D	×
C0034:15							
C0034:29	Illuminates	Not illuminate*2	RF ABS wheel-speed sensor/ABS sensor rotor	×	—	C, D	×
C0034:2F							
C0034:64							
C0036:07	Illuminates	Not illuminate*2	LR ABS sensor rotor	×	—	C, D	×
C0037:07			LR ABS wheel-speed sensor/ABS sensor rotor				
C0037:11	Illuminates	Not illuminate*2	LR ABS wheel-speed sensor	×	—	C, D	×
C0037:15							
C0037:29	Illuminates	Not illuminate*2	LR ABS wheel-speed sensor/ABS sensor rotor	×	—	C, D	×
C0037:2F							
C0037:64							
C0039:07	Illuminates	Not illuminate*2	RR ABS sensor rotor	×	—	C, D	×
C003A:07			RR ABS wheel-speed sensor/ABS sensor rotor				

DTC	ABS warning light	Brake system warning light (when parking brake is released)	Description	Fail-safe	Drive cycle	Self test type*1	Memory function
C003A:11	Illuminates	Not illuminate*2	RR ABS wheel-speed sensor	×	—	C, D	×
C003A:15							
C003A:29	Illuminates	Not illuminate*2	RR ABS wheel-speed sensor/ABS sensor rotor	×	—	C, D	×
C003A:2F							
C003A:64							
C0040:64	Not illuminate	Not illuminate	Brake switch	×	—	C, D	×
C0061:54	Not illuminate	Not illuminate	ABS HU/CM (unperformed initialization procedure)	×	—	C, D	×
C0062:28	Not illuminate	Not illuminate	SAS control module	×	—	C, D	×
C0062:54	Not illuminate	Not illuminate	ABS HU/CM (unperformed initialization procedure)	×	—	C, D	×
C0062:64	Not illuminate	Not illuminate	SAS control module	×	—	C, D	×
C0062:76	Not illuminate	Not illuminate	SAS control module	×	—	C, D	×
U0001:88	Illuminates	Not illuminate	CAN line	×	—	C, D	×
U0100:00	Illuminates*3	Not illuminate					
U0101:00	Not illuminate	Not illuminate					
U0151:00	Not illuminate	Not illuminate					
U0155:00	Illuminates*3	Not illuminate					
U0401:00	Not illuminate	Not illuminate	Signal error from PCM	×	—	C, D	×
U0402:00	Not illuminate	Not illuminate	Signal error from transmission/ transaxle	×	—	C, D	×
U0423:00	Not illuminate	Not illuminate	Signal error from instrument cluster	×	—	C, D	×
U0452:86	Not illuminate	Not illuminate	Signal error from SAS control module	×	—	C, D	×
U2101:00	Illuminates	Not illuminate	ABS HU/CM configuration	×	—	C, D	×
U2300:52							
U2300:54							
U2300:55							
U2300:56							
U2300:64							
U3000:49	Illuminates	Illuminates*3	ABS HU/CM internal malfunction	×	—	C, D	×
U3003:16	Illuminates	Not illuminate*4	Battery power supply	×	—	C, D	×
		Illuminates*5*6					
U3003:17	Illuminates	Illuminates					

*1 : C: CMDTC self test, D: ODDTC self test
*2 : Illuminates when two wheels or more have a malfunction
*3 : Not illuminate depending on malfunction content
*4 : When power supply voltage is 8.4—9.6 V for 0.7 s or more
*5 : When power supply voltage is 6.0—8.4 V for 0.8 s or more
*6 : When power supply voltage is 6 V or less for 0.05 s or more

Status byte for DTC

- The status byte is the two-digit code (two digits after hyphen (-)) after the seven-digit DTC.
- The status byte is a code which indicates the pending code, current/past malfunction status, or warning illumination status.
- The status byte is a code which indicates the pending code, current/past malfunction status, or warning illumination status.
- The status byte can be read by performing a CMDTC self test using the Mazda Modular Diagnostic System (M-MDS).
- For details on the status byte, refer to the explanation on the M-MDS when reading the DTC.



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Fail-safe function

- When the malfunction detection function determines a malfunction, the corresponding lights illuminates to alert the driver. At this time, the fail-safe function controls as indicated in the fail-safe function table.

Warning

- **If EBD control is prohibited, the rear wheels could lock-up before the front wheels. If this occurs, the vehicle could yaw and become unstable. Therefore, always inspect the system immediately if EBD control is prohibited.**

Fail-safe function table

×: Control enabled
—: Control disabled

DTC	ABS control	EBD control
C0010:01	—	—
C0011:01		
C0014:01		
C0015:01		
C0018:01		
C0019:01		
C001C:01		
C001D:01		
C0020:11	—	×
C0020:12		
C0020:13		
C0020:71		
C0030:07	—	×*1
C0031:07		
C0031:11		
C0031:15		
C0031:29		
C0031:2F		
C0031:64		
C0033:07	—	×*1
C0034:07		
C0034:11		
C0034:15		
C0034:29		
C0034:2F		
C0034:64		
C0036:07	—	×*1
C0037:07		
C0037:11		
C0037:15		
C0037:29		
C0037:2F		
C0037:64		
C0039:07	—	×*1
C003A:07		
C003A:11		
C003A:15		
C003A:29		
C003A:2F		
C003A:64		
C0040:64	×	×
C0061:54	×	×
C0061:64	×	×
C0062:28	×	×
C0062:54		
C0062:64		
C0062:76		
U0001:88	—	×

DTC	ABS control	EBD control
U0100:00	—*2	×
U0101:00	×	×
U0151:00	×	×
U0155:00	—*2	×
U0401:00	×	×
U0402:00	×	×
U0423:00	×	×
U0452:86	×	×
U2101:00	—	×
U2300:52		
U2300:54		
U2300:55		
U2300:56		
U2300:64		
U3000:49	—	—*2
U3003:16	—	×*3
		—*4*5
U3003:17	—	—

*1 : Control disabled when two wheels or more have a malfunction
*2 : Enabled depending on malfunction content
*3 : When power supply voltage is 8.4—9.6 V for 0.7 s or more
*4 : When power supply voltage is 6.0—8.4 V for 0.8 s or more
*5 : When power supply voltage is 6 V or less for 0.05 s or more

Snapshot data

- The snapshot data stores the currently detected DTC data.

—: Not applicable

Snapshot data item	Unit	Definition	Data read/use method	Corresponding data monitor items
AAT	°C, °F	Ambient air temperature	—	—
ABS	Inactive/Active	Antilock braking system	—	—
APP_STATUS	Accelerator Pedal Off/ Under 20%/ Over 20%/ FAIL	Accelerator pedal position	—	—
AYC	Inactive/Active	Active yaw control	—	—
BRK_F_P_R	Pa, psi	Brake fluid line hydraulic pressure (raw value)	—	—
BTCS	Inactive/Active	Brake traction control system	—	—
CFG_STATUS	Config Complete/ Not Configured/ Config Error	Configuration status	—	—
ECT_STATUS	Under 0 degrees C/ 0 - Under 80 degrees C/ Over 80 degrees C/ FAIL	Engine coolant temperature status	—	—
EDC	Inactive/Active	Engine drag control	—	—

Snapshot data item	Unit	Definition	Data read/use method	Corresponding data monitor items
IC_VPWR	V	Instrument cluster power supply	• The ABS HU/CM constantly receives the power supply voltage value of the instrument cluster sent via CAN signal from the instrument cluster. • If a DTC is detected, the ABS HU/CM records the power supply voltage of the instrument cluster when the DTC was detected, and it is displayed in the M-MDS.	VPWR*1
IG-ON_TIMER	hh:mm:ss*2	Elapsed time since ignition was switched ON **Note** • The instrument cluster records the elapsed time since the ignition was switched ON.	• The ABS HU/CM constantly receives the elapsed time since the ignition was switched ON sent via CAN signal from the instrument cluster. • If a DTC is detected, the ABS HU/CM records the elapsed time since the ignition was switched ON when the DTC was detected, and it is displayed in the M-MDS.	—
LAT_ACCL_R	G	Lateral acceleration (raw value)	—	LAT_ACCL_R
LON_ACCL_R	G	Longitudinal acceleration (raw value)	—	LON_ACCL_R
PMP_MT	Off/On	Pump motor	—	PMP_MT
PWR_MODE_KEY	Key Out/Key Recently Out/Key Approved (Position 0)/Post Accessory (Position 0)/ Accessory (Position 1)/ Post Ignition (Position 1)/Ignition On (Position 2)/Running (Position 2)/ Running - Starting In Progress (Position 2)/ Crank (Position 3)	• Key Out: Ignition switched off • Key Recently Out (Position 0): Elapsed time within 3 s since ignition was switched off • Accessory (Position 1): Ignition is switched to ACC • Post Ignition (Position 2): Elapsed time within 3 s since ignition was switched ON • Ignition On (Position 2): Ignition switched ON (engine off) • Running (Position 2): Ignition switched ON (engine on) • Running - Starting: Cranking condition	• The ABS HU/CM constantly receives the ignition switch status sent via CAN signal from the instrument cluster. • If a DTC is detected, the ABS HU/CM records the ignition switch status when the DTC was detected, and it is displayed in the M-MDS.	—

Snapshot data item	Unit	Definition	Data read/use method	Corresponding data monitor items
RPM_STATUS	Engine Stop/ Under 1500 rpm/ Over 1500 rpm/ FAIL	Engine RPM status	• The ABS HU/CM constantly receives the ignition switch status sent via CAN signal from the instrument cluster. • If a DTC is detected, the ABS HU/CM records the ignition switch status when the DTC was detected, and it is displayed in the M-MDS.	TACHOMTR*1
SHIFT_STATUS	P/N D/ R/ FAIL	Shift position status	• The ABS HU/CM constantly receives the selector lever position sent via CAN signal from the instrument cluster. • If a DTC is detected, the ABS HU/CM records the selector lever position when the DTC was detected, and it is displayed in the M-MDS.	—
STR_ANG_C	°	Steering wheel angle (calculated value)	—	—
TCS	Inactive/Active	Traction control system	—	—
TOTAL_DIST	km, ft, mi	Accumulated total traveled distance from completion of vehicle until ABS HU/CM detects DTC (Odometer value in instrument cluster)	The distance traveled when the ABS HU/CM detected a DTC can be calculated by performing the following procedure. 1. Verify the odometer value in the instrument cluster. 2. Verify the snap shot data item TOTAL_DIST. 3. Subtract 2 from 1.	—
TOTAL_TIME	hh:mm:ss*2	Accumulated total elapsed time since vehicle completion until ABS HU/CM detects a DTC **Note** • When the ROOM fuse is removed, and the ignition is switched off, the time is not included in the elapsed time.	The elapsed time when the ABS HU/CM detected a DTC can be calculated by performing the following procedure. 1. Verify the PID item TOTAL_TIME of the instrument cluster. 2. Verify the snap shot data item TOTAL_TIME. 3. Subtract 2 from 1.	TOTAL_TIME*1
VPWR	V	Power supply	—	VPWR_B_SOL
VSPD	KPH, MPH	Vehicle speed	—	VSPD

Snapshot data item	Unit	Definition	Data read/use method	Corresponding data monitor items
VSPD_STATUS	Stop/ 0 - 10 km/h/ Over 10 km/h/ FAIL	Vehicle speed status	• The ABS HU/CM constantly receives the vehicle speed sent via CAN signal from the instrument cluster. • If a DTC is detected, the ABS HU/CM records the vehicle speed when the DTC was detected, and it is displayed in the M-MDS.	SPEEDOMTR*1
YAW_RATE_R	°/s	Yaw rate (raw value)	—	YAW_RATE_R

*1 : PID for instrument cluster
*2 : Fractions of seconds may be indicated after the decimal point

PID/data monitor function

- The PID/data monitor function is used for optionally selecting input/output signal monitor items preset in the ABS HU/CM and reading them out in real-time.

Monitor item (Mazda Modular Diagnostic System (M-MDS) display)	Definition	Unit/operation
BRAKE_SW	Brake switch signal	Off/On
LAT_ACCL_C	Lateral-G value (calculated value)	G
LAT_ACCL_R	Lateral-G value (raw value)	
LON_ACCL_C	Longitudinal-G value (calculated value)	G
LON_ACCL_R	Longitudinal-G value (raw value)	
P_BRAKE_SW	Parking brake switch signal	Off/On
PMP_MT	**Note** • Displays in the M-MDS but it does not operate.	
PMP_MT_SP		
V_INLET_LF	LF inlet solenoid valve status	Off/On
V_INLET_LR	LR inlet solenoid valve status	
V_INLET_RF	RF inlet solenoid valve status	
V_INLET_RR	RR inlet solenoid valve status	
V_OUTLET_LF	LF outlet solenoid valve status	
V_OUTLET_LR	LR outlet solenoid valve status	
V_OUTLET_RF	RF outlet solenoid valve status	
V_OUTLET_RR	RR outlet solenoid valve status	
VPWR_B_SOL	Power supply	V
VSPD	Vehicle speed	KPH, MPH
WSPD_SEN_LF	LF ABS wheel-speed sensor signal	KPH, MPH
WSPD_SEN_LR	LR ABS wheel-speed sensor signal	
WSPD_SEN_RF	RF ABS wheel-speed sensor signal	
WSPD_SEN_RR	RR ABS wheel-speed sensor signal	
YAW_RATE_C	Yaw rate value (calculated value)	°/s
YAW_RATE_R	Yaw rate value (raw value)	

Simulation function

- The simulation function is used for optionally selecting simulation items of input/output parts preset in the ABS HU/CM, and to operate them regardless of CM control.
- To protect the hydraulic unit interior, operate output related parts for only 2 s or less when using the simulation function.

Simulation item	Output part	Operation	Operating condition
PMP_MT_SP	Pump motor	Off/On	Switch the ignition ON
V_INLET_LF	LF inlet solenoid valve		
V_INLET_LR	LR inlet solenoid valve		
V_INLET_RF	RF inlet solenoid valve		
V_INLET_RR	RR inlet solenoid valve		
V_OUTLET_LF	LF outlet solenoid valve		
V_OUTLET_LR	LR outlet solenoid valve		
V_OUTLET_RF	RF outlet solenoid valve		
V_OUTLET_RR	RR outlet solenoid valve		

External tester communication function

- The external tester communication function enables communication of diagnostic data (DTC read-outs, input/output signal read-outs, operation of input/output parts) between the ABS HU/CM and an external tester.
- By connecting the Mazda modular diagnostic system (M-MDS) to the DLC-2, diagnostic data can be sent and received between the Mazda modular diagnostic system (M-MDS) and the ABS HU/CM via CAN lines.
- The ABS HU/CM receives the command signals of the malfunction detection function, PID/data monitor function, and simulation function from the Mazda modular diagnostic system (M-MDS), and sends DTCs and data regarding the operating status of each input/output part to the Mazda modular diagnostic system (M-MDS).

Diagnostic function name	Signal received	Signal sent
Malfunction detection function	DTC verification signal	DTC(s)
PID/data monitor function	Command signal to read selected monitor item	Monitored data for requested monitor item
Simulation function	Operation command signal for selected simulation item	Output part drive signal